# Workbook: Muscle Microanatomy

Week 4 workbook · BIO 004 Human Anatomy · Dr. Sharilyn Rennie · Solano Community College

A comprehensive workbook on the microanatomy of skeletal muscle: from whole muscle down to the sarcomere and its proteins, the connective tissue wrappings, the neuromuscular junction, and a comparison of skeletal, cardiac, and smooth muscle.

**How to use this workbook:** Read each question carefully and write your answer in the space provided. Work without looking up the answer first. Once you have written your answer, check it against the online workbook by clicking *Reveal* on that question. Star or circle anything you missed and come back to it later in the week.

## Levels of organization · 3 questions

**Q01 · DOK 1 · Recall**

Name the levels of skeletal muscle organization from whole muscle down to the contractile protein.

**Q02 · DOK 1 · Recall**

Name the three connective tissue wrappings of skeletal muscle and what each surrounds.

**Q03 · DOK 1 · Recall**

Distinguish tendon from aponeurosis.

## Muscle fiber anatomy · 4 questions

**Q04 · DOK 1 · Recall**
List the distinctive features of a skeletal muscle fiber.

**Q05 · DOK 1 · Recall**
What is the sarcolemma, and what is unusual about it compared to a typical cell membrane?

**Q06 · DOK 1 · Recall**
What is the sarcoplasmic reticulum, and what does it store?

**Q07 · DOK 1 · Recall**
What is a triad in a muscle fiber, and what does it accomplish?

## Sarcomere and myofilaments · 5 questions

**Q08 · DOK 1 · Recall**
What is a sarcomere, and what defines its boundaries?

**Q09 · DOK 1 · Recall**
Compare thick and thin filaments by protein and where in the sarcomere each sits.

**Q10 · DOK 1 · Recall**

Name the bands and lines of a sarcomere and what each one represents.

**Q11 · DOK 1 · Recall**

What are tropomyosin and troponin, and what do they do?

**Q12 · DOK 2 · Apply**

What is the basic idea of the sliding filament theory, expressed without molecular detail?

## Neuromuscular junction · 3 questions

**Q13 · DOK 1 · Recall**

What is the neuromuscular junction, and what are its three main components?

**Q14 · DOK 1 · Recall**

What is a motor unit?

**Q15 · DOK 1 · Recall**

What neurotransmitter is released at the NMJ, and what receptor binds it on the muscle side?

## Three muscle types compared · 2 questions

**Q16 · DOK 2 · Compare**

Compare skeletal, cardiac, and smooth muscle by striations, number of nuclei per cell, control, and one distinctive feature of each.

**Q17 · DOK 2 · Apply**

Smooth muscle contracts even though it has no sarcomeres. What arrangement of contractile proteins makes that possible?